



# DEVELOPMENT & ERUPTION OF THE TEETH

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# SUMMARY

- Tooth development begins IU life
- Primary dentition complete by 3 yrs
- Mixed dentition critical stage
- Permanent dentition completes by adulthood
- Dental age useful indicator



- A mature neuromuscular-controlled movement of the mandible requires the presence and articulation of the teeth and proprioceptive input from the periodontium. Thus the contact of opposing first primary molars is the beginning of the development of occlusion and a neuromuscular substrate for more complex mandibular and tongue functions





**FIG. 2.6** Neonatal skull showing large brain case and orbit; the neurocranium is larger than the splanchnocranium, which contains the jaws and all the developing teeth.

From Avery JK, Chiego DJ Jr: *Essentials of oral histology and embryology*, ed 3, St. Louis, 2006, Mosby.

# Beginning of Tooth Development

- Starts at **6th week intrauterine life**
- Jaws small compared to cranium
- Mandibular arch larger than maxillary
- All stages of tooth formation present
- The shape of the prenatal head varies considerably, but the relative difference among the brain case, orbits, and lower face height remains the same.
- All stages of tooth formation fill both jaws during this stage of development

The first indication of tooth formation occurs as early as the sixth week of prenatal life, when the jaws have assumed their initial shape

The jaws are rather small compared with the large brain case and orbits.

The lower face height is small compared with the neurocranium

the vertical dimensions of the jaws are only minimally developed.

When the jaws close at this stage in the development of the dentition, they make contact with the tongue, which in turn makes contact with the cheeks.

# The Dentitions

- The human dentitions are usually categorized as being primary, mixed (transitional), and permanent dentition

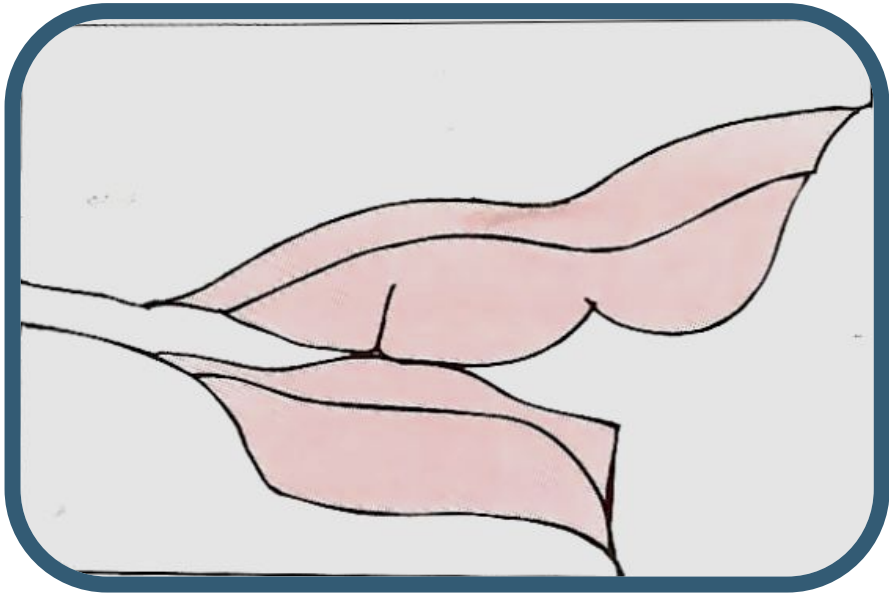


# Neonatal Skull Characteristics

- Large neurocranium
- Small face height
- Minimal vertical jaw development
- Tongue fills oral cavity
- Clinical note: jaws grow postnatally to accommodate teeth



# Neonatal period last upto six months



# DEVELOPMENT OF PRIMARY DENTITION

- **At Birth**
- Usually no erupted teeth
- Rare: natal teeth (mandibular incisors)
- Tooth development continues after birth
- Alveolar bone height increases
- Anterior jaw growth limited after 1 year
- Alveolar bone develops with tooth eruption

Considerable growth follows birth in the neurocranium and splanchnocranium. Usually at birth, no teeth are visible in the mouth; occasionally, however, infants are born with erupted mandibular incisors. Development of both primary and permanent teeth continues in this period, and jaw growth follows the need for additional space posteriorly for additional teeth. In addition, the alveolar bone height increases to accommodate the increasing length of the teeth. However, growth of the anterior parts of the jaws is limited after about the first year of postnatal life.

# Primary Dentition



# Sequence of Primary Tooth Eruption

- **A B D C E sequence:**

- Central incisor
- Lateral incisor
- First molar
- Canine
- Second molar

# Primary Eruption Timeline

- 6–10 months: mandibular central incisors
- 1 year: most incisors
- 16 months: first molars
- 18–20 months: canines
- 24–30 months: second molars
- Complete by ~30 months



# Root Formation & Eruption

- Tooth emerges when  $\sim 3/4$  root formed
- Occlusion begins when first molars contact
- Important for:
  - Neuromuscular development
  - Mastication
  - Speech

# Neuromuscular Development

- Occlusion allows:
- Controlled mandibular movement
- Tongue coordination
- Functional chewing
- First molar contact → start of functional occlusion



# PRIMARY DENTITION PERIOD

- Completed by 30 months
- Stable until ~6 years
- 20 teeth present



# Arch Changes in Primary Dentition

- Arch width mostly stable
- Slight intercanine increase
- Sagittal jaw growth continues
- Vertical face height increases

# Importance of Primary Teeth

- Functions:
- Maintain arch space
- Guide permanent teeth
- Speech development
- Mastication

## ***Premature loss leads to:***

- Crowding
- Malocclusion
- Space loss

# Mixed Dentition Period

- Age: **6–12 years**
- Begins:
- Permanent first molar eruption
- Incisor exchange
- Ends:
- Loss of last primary tooth





## Intertransitional period

Age

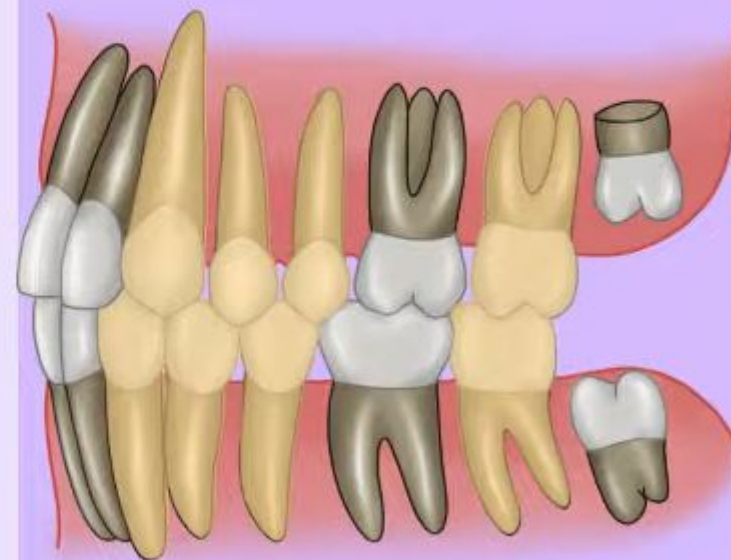
8-9 years



## Second transitional period

Age

9-10 years

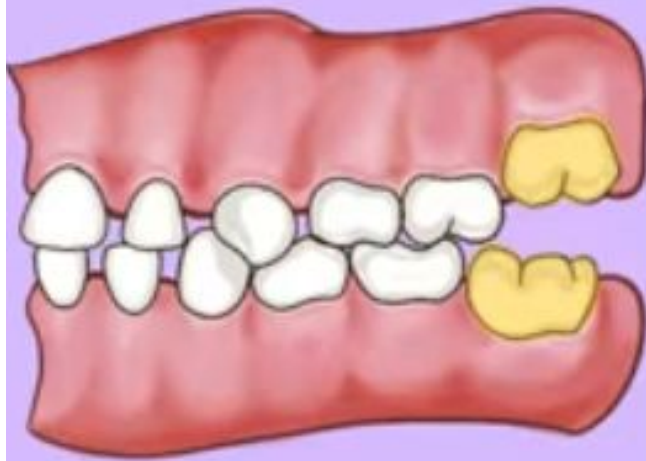


# First Transition Phase

- 6-year molars erupt
- Primary incisors shed
- Permanent incisors erupt
- Duration: ~2 years

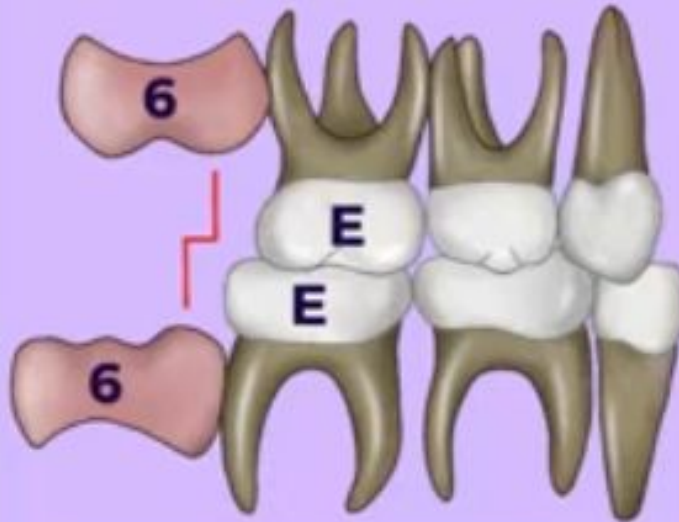


**1**



**MORE VIDEOS** of first  
permanent molar  
before eruption

**2**



**Relationship between  
permanent first molar  
and primary second  
molars**

**3**

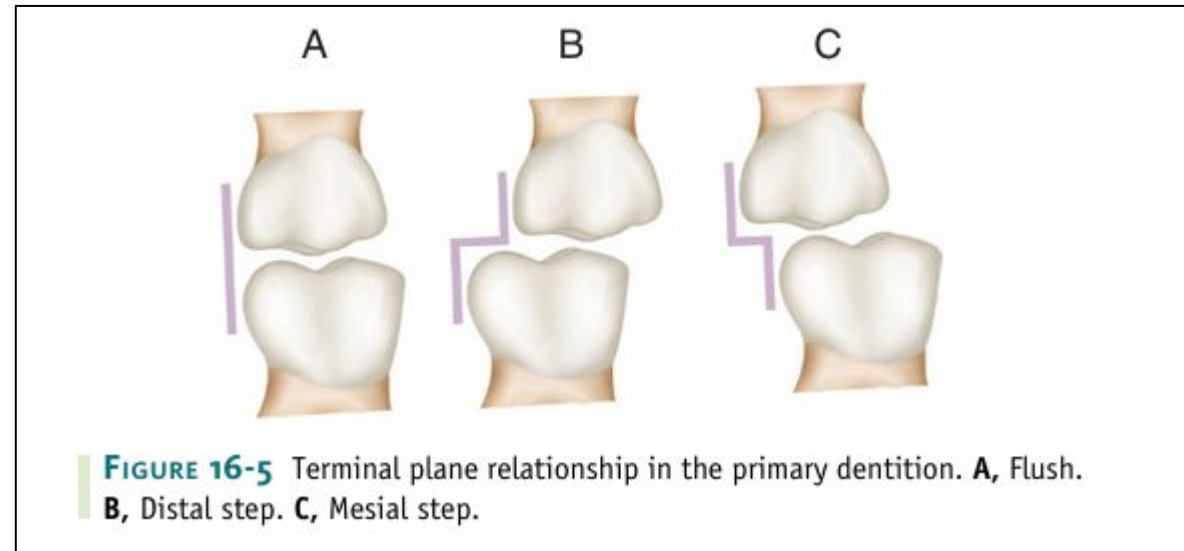


**Relation between the  
maxilla and mandible  
in a sagittal plane**

# Guidance of Eruption

- First molar guided by distal of primary molar
- Distal step → malocclusion risk
- Important orthodontic stage

The permanent teeth do not begin eruptive movements until after the crown is completed. During eruption, the permanent mandibular first molar is guided by the distal surface of the second primary molar. If a distal step in the terminal plane is evident, malocclusion occurs



# Premature Loss of Primary Teeth

- Causes:
- Caries
- Trauma
- Consequences:
- Space loss
- Crowding
- Malocclusion
- Primary teeth remain functional up to 12 years

the primary dentition may be in use from age 2 to 7 or older, or approximately 5 or more years in all. Some of the teeth are in use from 6 months until 12 years of age, or 11.5 years in all. Premature loss of primary teeth, retention of primary teeth, congenital absence of teeth, dental anomalies, and insufficient space are considered important factors in the initiation and development of an abnormal occlusion. Premature loss of primary teeth from dental neglect is likely to cause a loss of arch length with a consequent tendency for crowding of the permanent dentition



## EARLY PRIMARY TOOTH LOSS

the consequences of no space maintainers



poor eruption of adult teeth



impacted adult teeth

# PERMANENT DENTITION

- **Permanent Teeth**

- Total: 32 teeth Complete by 18–25 years

Includes:

- Incisors
- Canines
- Premolars
- Molars

There are four or more centers of formation (developmental lobes) for each tooth.

The formation of each center proceeds until a coalescence of all of them takes place., signs are found on the surfaces of the crowns and roots; these are called developmental grooves

Fractures of the teeth occur most commonly along these grooves .

The follicles of the developing incisors and canines are in a position lingual to the deciduous roots The developing premolars, which eventually take the place of deciduous molars, are within the bifurcation of primary molar roots







As with the first molar, at age 6 years, 50% of individuals have reached the the age of emergence for the central incisor.

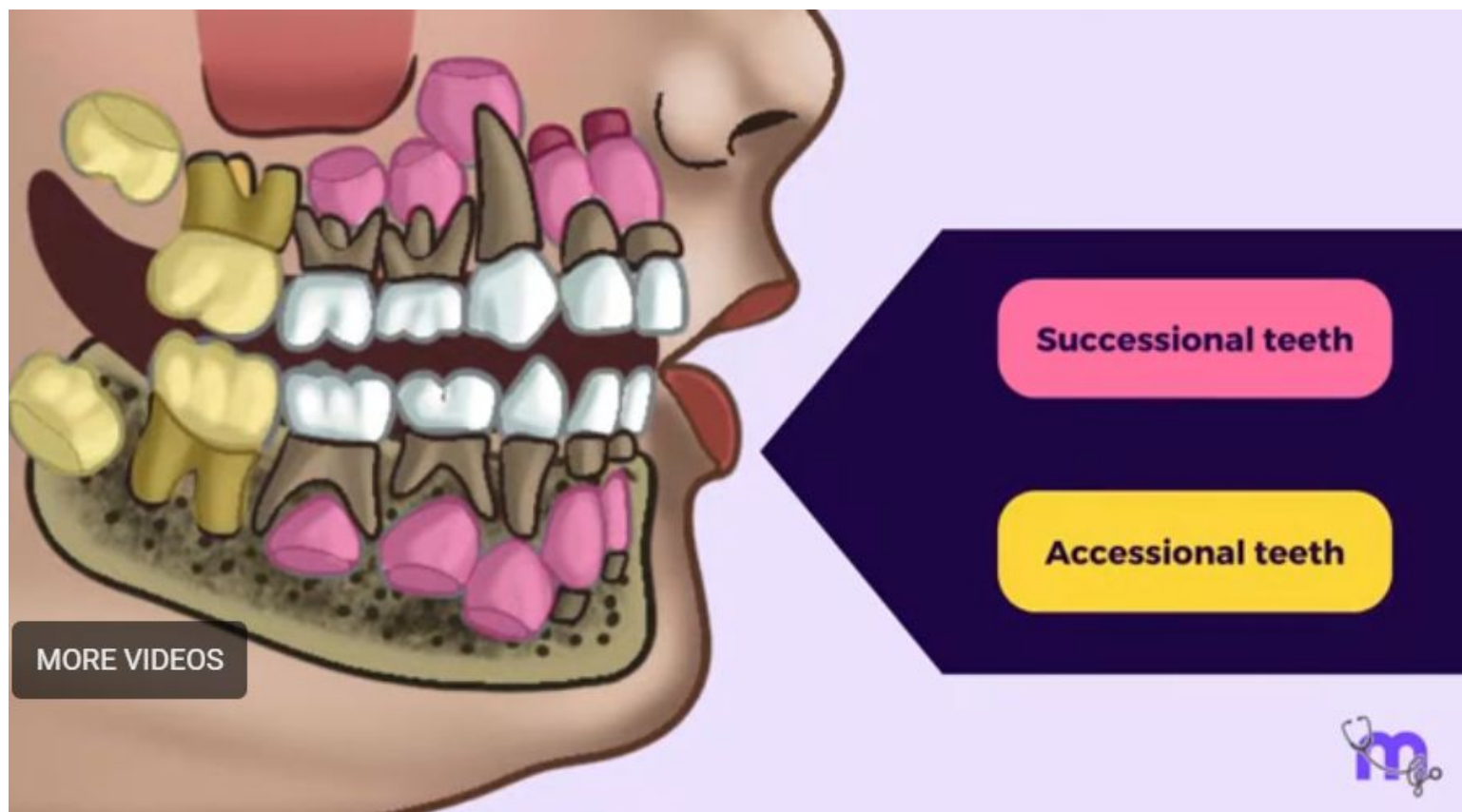
The mandibular permanent teeth tend to erupt before maxillary teeth. The mandibular central incisor usually erupts before the maxillary central incisor and may erupt simultaneously with or even before the mandibular

Before the permanent central incisor can come into position, the primary central incisor must be exfoliated.

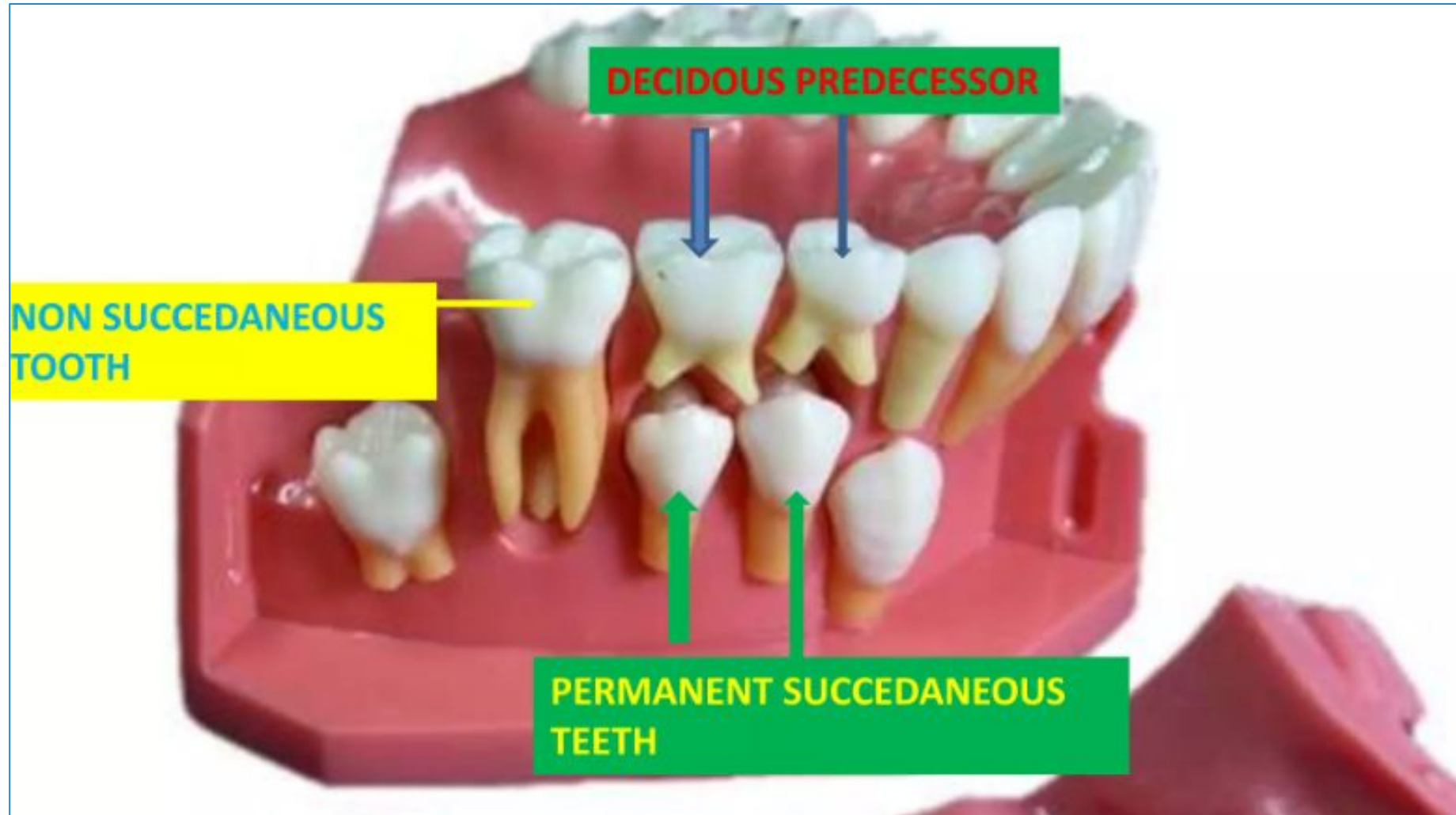
This occurs through the resorption of the deciduous roots. The permanent tooth in its follicle attempts to move into the position held by its predecessor. Its influence on the primary root evidently causes resorption of the root, which continues until the primary crown has lost its anchorage, becomes loose, and is finally exfoliated.



**FIG. 2.11** Eruption of the permanent central incisor. Note the incisal edges demonstrating mamelons and the width of the emerging incisors.



# Succedaneous Teeth



teeth

# Permanent Eruption Sequence

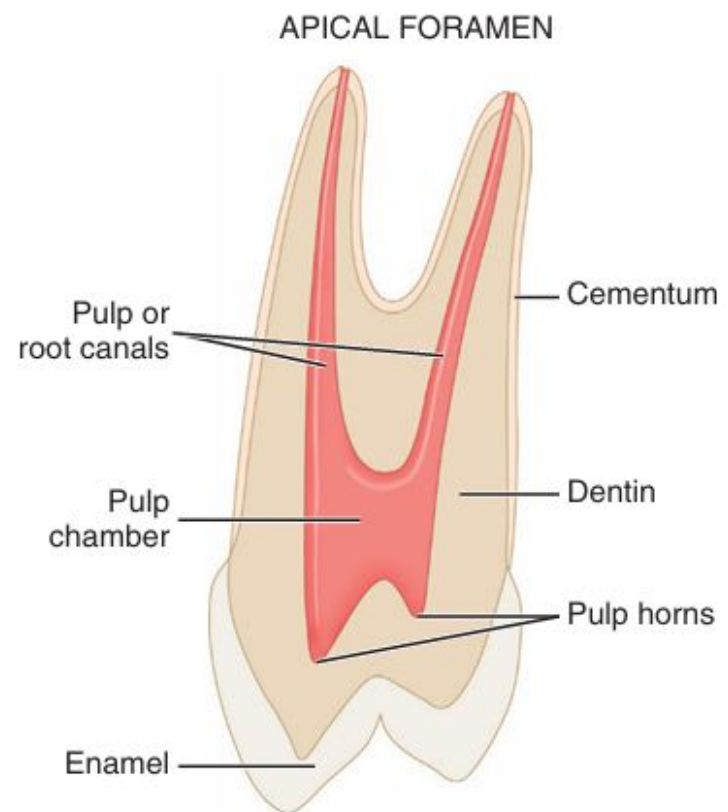
- 6 yrs: first molar
- 6–7 yrs: central incisors
- 7–8 yrs: lateral incisors
- 9–11 yrs: premolars
- 10–12 yrs: canines
- 12 yrs: second molars
- 17–25 yrs: third molars



# Tooth Size Determinants

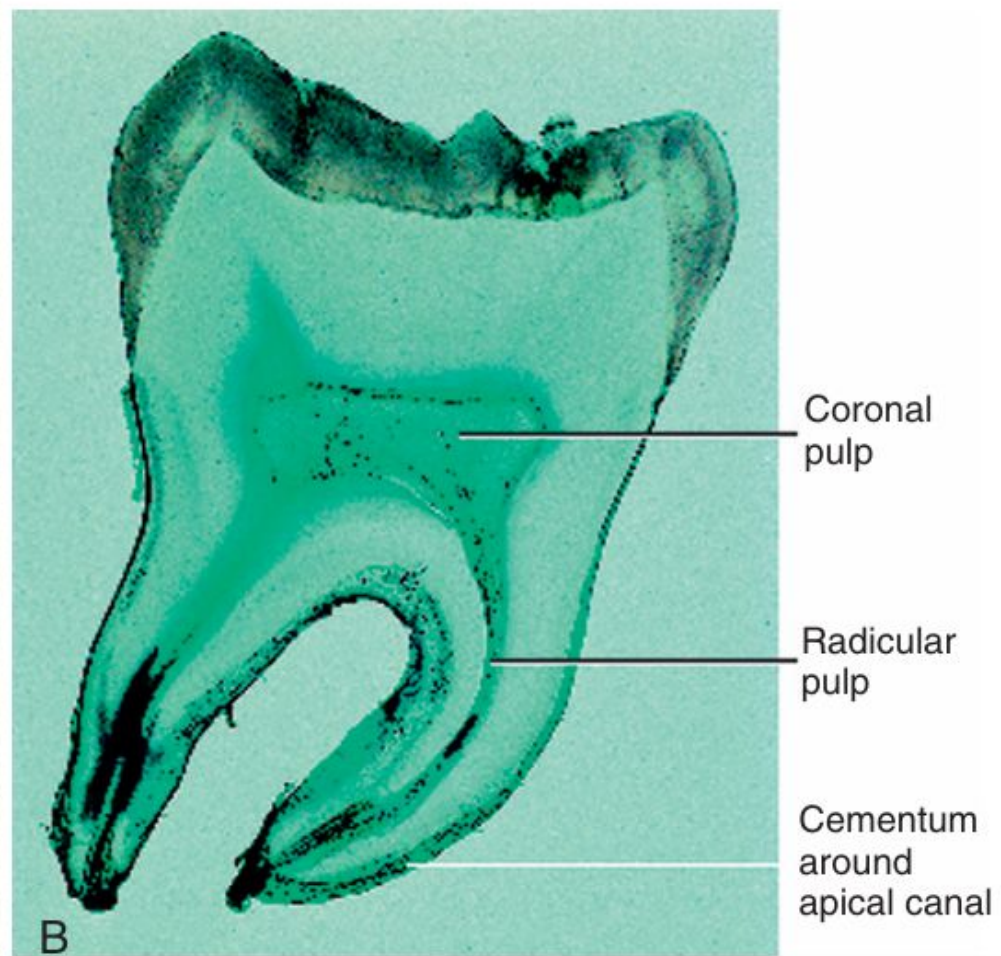
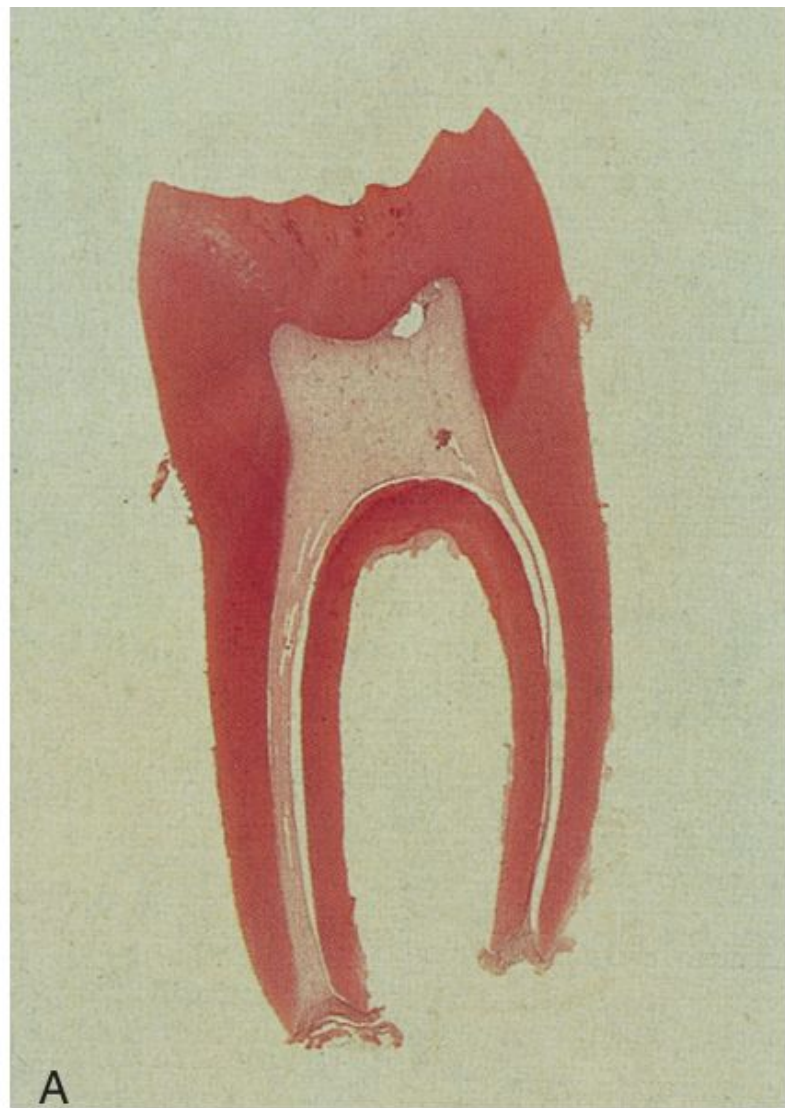
- Mostly genetic
- Gender difference ~4%
- Tooth–arch size discrepancy → crowding
- Lapps, a population with perhaps the smallest teeth, and the Australian aborigines, with perhaps the largest teeth.





**FIGURE 13-1** Buccolingual section of a maxillary premolar.





# Dental Pulp

Connective Tissue. It contains

- Blood vessels
- Nerves
- Lymphatics

Functions:

- Forms dentin
- Sensory
- Defensive
- Large in young teeth → more sensitivity

connective tissue organ

When the tooth is newly erupted, the dental pulp is large; it becomes progressively smaller as the tooth is completed.

The pulp is relatively large in primary teeth as well as in young permanent teeth .

The teeth of children and young people are more sensitive than the teeth of older people to thermal change and dental operative procedures (heat generation).

The opening of the pulp cavity at the apex is constricted and is called the apical foramen.

The pulp keeps its tissue forming function (e.g., to form secondary dentin), especially with the advance of dental caries toward the pulp.

The pulp cavity becomes smaller and more constricted with age

The pulp chamber within the crown may become almost obliterated with a secondary deposit (e.g., osteodentin). This process is not as extensive in deciduous teeth

# CEMENTOENAMEL JUNCTION

- CEJ
- Junction of enamel & cementum Important landmark for:
  - Periodontal probing
  - Restorations
  - Sensitivity
- **Types of CEJ Relationship**
  1. Enamel overlaps cementum
  2. Edge-to-edge
  3. Gap (dentin exposed)
  4. Cementum overlaps enamel
- Clinical relevance in recession & caries

# DENTAL AGE

- **Dental Age Concept**
- Best indicator of physiologic age in children Based on:
  - Tooth formation
  - Eruption
  - Root development
- **Methods to Assess Dental Age**
- Radiographs
- Eruption sequence
- Root resorption
- Crown formation stages
- More accurate in children than adults

# Environmental Influences

- Tetracycline → discoloration
- Fluoride → fluorosis (excess)
- Nutrition → minor effect
- Tooth formation less affected than eruption

# FAQ??

1. First tooth to erupt?
2. Age of mixed dentition?
3. Succedaneous teeth?
4. Importance of primary teeth?
5. CEJ clinical relevance?